Are Medical Vision-Language Models Better Than Their Base Models? A Paired Zero-Shot Evaluation on Three Medical Imaging Tasks

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Abstract

Medical vision-language models (VLMs) are usually built by further training a general-purpose VLM, and are expected to beat that base model on medical images. We test this directly with two small open model pairs: MedVLM-R1 versus its base Qwen2-VL-2B, and MedGemma 4B versus its base Gemma 3 4B. All four models answer the same items, zero-shot, with the same prompts, on three tasks: benign/malignant classification of breast ultrasound (BUS-BRA, 400 images), closed-ended CT/MRI question answering (RadImageNet-VQA, 500 questions) and open-ended medical question answering in Bangla (BanglaMedVQA, 500 questions). Medical adaptation did not give a consistent gain. MedVLM-R1 answered almost exactly like its base model on all three tasks. MedGemma improved on its base model only in Bangla (+7.4 LAVE points), was not significantly better on breast ultrasound, and was slightly worse on RadImageNet-VQA. All models recognised modality and anatomy far better than disease, and often answered from fixed preferences (one label, "yes" or "no", one option letter). In Bangla, most answers were in English, and the LLM-judged metric scored correct English answers inconsistently.
Contribution of the Paper: A paired comparison of medical VLMs against their own base models on three tasks, showing that medical adaptation does not consistently improve zero-shot performance and that answer biases and cross-lingual scoring can hide this behind headline accuracy.

Keywords: medical vision-language models, zero-shot evaluation, medical visual question answering, breast ultrasound, Bangla, answer bias

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1. INTRODUCTION

A growing number of vision-language models (VLMs) are released for medicine. Most of them are not trained from scratch: they take a general-purpose VLM and train it further on medical images and text, through instruction tuning [1, 2], large-scale medical training [3] or reinforcement learning on a small set of medical questions [4]. The expectation is that the medical model should be better than the model it started from.

This is rarely tested directly. New medical models are usually compared with other medical models or with large proprietary systems, and the most informative baseline, the original base model under the same conditions, is often missing. When the comparison has been made carefully, medical models have often not come out ahead [5], and some benchmarks found medical VLMs scoring below general ones [6].

We ask this question for small, open VLMs that a hospital or research group with limited resources could run locally: *when used zero-shot, does a medical VLM answer medical image questions better than its own base model?* We compare two pairs. MedVLM-R1 [4] was built from Qwen2-VL-2B [7] with reinforcement learning, and MedGemma 4B [3] was built from Gemma 3 4B [8] with large-scale medical training. Within each pair the architecture is the same, so differences can be attributed to the medical adaptation.

We use three tasks, each adding something the previous one cannot show:

- **Breast ultrasound classification** on BUS-BRA [9]:

one binary diagnostic decision with a known chance level. It shows whether a model uses the image at all.

- **Closed-ended CT/MRI VQA** on RadImageNet-VQA [10]: yes/no and multiple-choice questions that separate recognising anatomy from recognising disease, scored exactly.
- **Open-ended Bangla VQA** on BanglaMedVQA [11]: free-text answers to questions in a low-resource language, closest to how a clinician or patient might actually use a model.

Across the three tasks, medical adaptation did not give a consistent advantage: one medical model behaved almost as a copy of its base model, and the other was better on one task, not significantly different on one and slightly worse on one. The answers also show *why* accuracy alone is misleading here: several models gave nearly the same answer to every image, and the Bangla metrics mis-scored correct answers given in English.

Our contributions are: (1) a paired, base-model-controlled zero-shot comparison of two medical VLMs on three tasks with identical items, prompts and decoding; (2) an analysis of how the models fail, including constant answers, yes/no and option-letter preferences, and the gap between anatomy and disease questions; and (3) an audit of the Bangla evaluation showing that an LLM-judged metric [12] gives inconsistent credit to correct English answers.

2. RELATED WORKS

2.1. Medical VLMs and their base models

Base models. Qwen2-VL [7] combines a vision transformer with the Qwen2 language model. It processes images at their native resolution, turning each image into a variable number of visual tokens, and was released at 2B, 8B and 72B parameters. Gemma 3 [8] is a family of open models from 1B to 27B parameters; the report introduces vision input through a SigLIP-based encoder, a longer context and improved multilingual ability over Gemma 2. We use the instruction-tuned 2B and 4B models.

Medical VLMs. Early medical VLMs adapted general models with biomedical image–text data: LLaVA-Med [1] fine-tuned LLaVA on PubMed Central figure–caption pairs and GPT-4-generated instructions, Med-PaLM M [13] trained one large model on 14 biomedical tasks, and HuatuoGPT-Vision [2] was trained on 1.3 million medical VQA samples refined with GPT-4V. The two medical models we test represent two different strategies for small models. **MedGemma** [3] builds on Gemma 3 with a medically tuned vision encoder (MedSigLIP) and further medical image and text training; its report describes gains of 2.6–10% on out-of-distribution medical multimodal question answering over the base models. **MedVLM-R1** [4] applies Group Relative Policy Optimization [14] to Qwen2-VL-2B using only 600 MRI multiple-choice questions. The reward checks that the model writes its reasoning in `<think>` tags and its final letter in `<answer>` tags, and that the letter is correct. The authors report an accuracy increase from 55.11% to 78.22% on MRI, CT and X-ray test questions.

2.2. Evaluations of medical VLMs

VQA-RAD [15] and SLAKE [16] are small radiology VQA sets often used for fine-tuning. Larger evaluation benchmarks have reported limits of current models. OmniMedVQA [6] collected questions from 73 datasets and found that medical-specialised VLMs performed worse than general ones. GMAI-MMBench [17] evaluated 50 VLMs and the best, GPT-4o, reached 53.96%. ProbMed [18] added negated questions with hallucinated findings and found that strong models performed worse than random on specialised diagnostic questions. A case study of GPT-4V [19] found good recognition of modality and anatomy but poor diagnosis. RadImageNet-VQA [10], one of our datasets, reported that VLMs struggle with fine-grained pathology and that performance falls to near random without the image.

Closest to our work, Jeong et al. [5] compared medical language and vision-language models head-to-head with their base models, accounting for statistical uncertainty, and found that the medical VLMs they tested did not consistently improve over their bases in zero- and few-shot settings. We apply the same paired logic to newer medical VLMs, to classification and VQA tasks including a low-resource language, and add an analysis of answer behaviour.

2.3. Multilingual medical VQA and answer biases

Medical VQA has been evaluated almost only in English. IndoRad-VQA [20] translated VQA-RAD into Indonesian and found gaps of 8–25% between English and Indonesian, including answers in the wrong language. BanglaMedVQA [11], the first Bangla medical VQA dataset, evaluated nine VLMs and found that medical VLMs (Med-LLaVA and MedGemma) underperformed general-purpose ones in zero-shot Bangla. Two known biases matter for our analysis: VQA models can exploit answer priors such as favouring "yes" [21], and language models prefer certain option letters in multiple-choice questions [22].

3. METHODOLOGY

3.1. Study design

The study has one controlled comparison: each medical VLM against its own base model (Fig. 1, Table 1). For each task, all four models receive the same items in the same order, with the same prompt and the same deterministic decoding; only the model-loading code differs. No

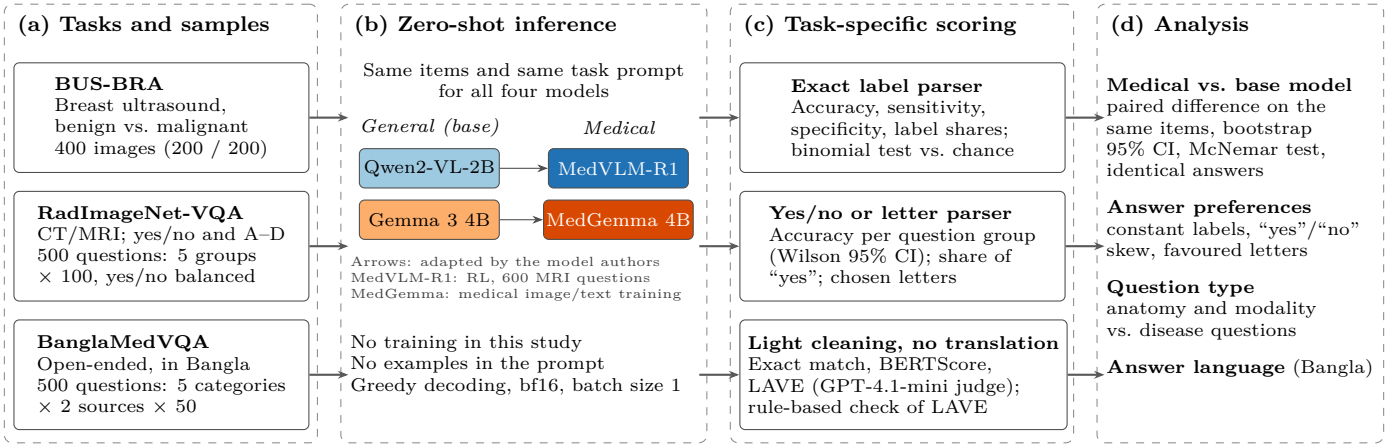


Figure 1: Experimental pipeline. (a) Each task uses a fixed, seeded sample that is identical for all models. (b) The four models answer every item zero-shot with the same task prompt; the medical models were adapted from their base models by their original authors, and no model was trained in this study. (c) Answers are scored with task-specific parsers and metrics. (d) Each medical model is compared with its own base model on the same items, and the answers are analysed for fixed preferences and by question type.

Table 1: Model pairs. Each medical model was trained from the general model in the same row (Hugging Face IDs in typewriter font).

	General (base)	Medical version
∼2B	Qwen2-VL-2B-Instruct [7] `Qwen/Qwen2-VL-2B-Instruct`	MedVLM-R1 [4] `JZPeterPan/MedVLM-R1` RL (GRPO) on 600 MRI questions
∼4B	Gemma 3 4B-it [8] `google/gemma-3-4b-it`	MedGemma 4B-it [3] `google/medgemma-4b-it` medical image and text training



(a) BUS-BRA, benign (b) BUS-BRA, malignant (c) BanglaMedVQA

Figure 2: Example images from our samples. (a, b) BUS-BRA [9] images bus_0007-l and bus_0001-l (CC BY 4.0); only MedGemma classified both correctly. (c) ChestX-ray8 [23] image 00001170_046 with the question কোনও অস্বাভাবিকতা আছে কি? ("Are any abnormalities present?"; reference হ্যাঁ, "yes"). RadImageNet-VQA images are not shown because its research-use agreement does not allow any part of the dataset to be reproduced.

model is fine-tuned and the prompts contain no examples. We chose the zero-shot setting because it is how a team without labelled data or fine-tuning resources would use these models, and because it measures what the medical adaptation itself adds. Using one prompt for both members of a pair keeps the comparison fair, at the cost of not giving each model its own best prompt (Section 5).

3.2. Datasets and sampling

Fig. 1(a) summarises the three samples and Fig. 2 shows example images. All samples were drawn with seed 42 and saved, so every model saw identical items. Every binary label set is balanced (benign/malignant, and "yes"/"no" within each yes/no group), so a model that always gives the same answer scores exactly chance and any answer preference becomes visible.

BUS-BRA [9] contains 1,875 breast ultrasound images from 1,064 patients with biopsy-proven benign or malignant labels (1,268 benign, 607 malignant). We chose it because the task is a clinically meaningful binary decision with a clear chance level, the labels are biopsy-proven, the dataset is openly licensed (CC BY 4.0), and case identifiers let us take one image per case. We sampled 400 images from 400 different cases, excluding 20 cases used in an earlier pilot run and 13 cases flagged during data checks. The final ID list was frozen and verified with a SHA-256 hash before each run.

RadImageNet-VQA [10] is built from the CT and MRI part of RadImageNet [24]. Its benchmark split has 9,000 questions on 1,000 images from eight anatomical regions. We kept the 7,000 yes/no and multiple-choice questions, so that every answer could be scored exactly, and grouped them by format and topic into five groups (Table 2): anatomy and pathology multiple choice, and anatomy, abnormality and pathology yes/no. We drew 100 questions per group.

BanglaMedVQA [11] has 7,000 Bangla questions, half on chest X-rays from ChestX-ray8 [23] and half on figures from biomedical papers collected in MedICaT [25]. The questions fall into five categories: *modality*, *organ*, *abnormality* (yes/no), *condition* and *position*. The question–answer pairs were generated from image metadata with Gemini and translated into Bangla; two medical specialists accepted 97.2% and 97.6% of a 1,000-pair subset [11]. We drew 50 questions for each source–category cell from

the training splits (3,500 questions per source), at most one per image within a cell. The original paper reports results on the test splits (200 questions per source), so our numbers are not directly comparable with theirs. No model in our study was trained on BanglaMedVQA.

3.3. Prompts and question types

Each task used one fixed prompt for all models (Table 2), which was not changed once evaluation started. For BanglaMedVQA we used the dataset authors' published zero-shot system prompt verbatim and passed the Bangla question as the user message, to stay as close as possible to the original benchmark. Table 2 also gives one real question from our samples for each of the ten question types. The RadImageNet-VQA groups separate *where* the image is from (anatomy) from *what is wrong* (abnormality, pathology); BanglaMedVQA makes a similar split between *general* (modality, organ) and *specialised* (abnormality, condition, position) categories [11].

3.4. Inference and answer parsing

All models ran in bfloat16 with greedy decoding (no sampling), batch size 1 and seed 42, using each model's own chat template. Greedy decoding makes outputs deterministic for a given setup, and batch size 1 avoids padding effects. The output limit was 32 new tokens for BUS-BRA, 16 for RadImageNet-VQA and 96 for BanglaMedVQA (256 for MedVLM-R1, so that a reasoning block, if produced, would not cut off the answer). The BUS-BRA runs pinned model revisions and library versions and ran on Kaggle T4 GPUs. In the two VQA runs, MedVLM-R1's own processor did not load with the installed `transformers` version, so we used the processor of its base model, Qwen2-VL-2B-Instruct; for BanglaMedVQA the Qwen processors were limited to 256–1280 visual patches of 28×28 pixels.

Answers that could not be parsed would count as wrong, but every output was parseable. BUS-BRA answers had to be exactly `benign` or `malignant`; RadImageNet-VQA took the first standalone letter A–D or the first "yes"/"no"; BanglaMedVQA answers were only stripped of whitespace, quotes and a final full stop or Bangla *dari*. All 3,600 classification and closed-VQA outputs were a single word or letter, and no model produced `<think>` or `<answer>` tags in any task.

3.5. Metrics and statistics

For BUS-BRA and RadImageNet-VQA we report accuracy with 95% Wilson intervals [26] and a one-sided exact binomial test against chance. For BUS-BRA we add sensitivity and specificity, and for yes/no questions the share of "yes" answers, since a balanced set makes any answer preference visible.

For BanglaMedVQA we follow the original paper and report exact match (EM), BERTScore F1 [27] (cased multilingual BERT-base, layer 9, no rescaling) and LAVE [12], in which an LLM judge rates each answer between 0 and 1 against the reference. As in the original paper, the judge was GPT-4.1-mini with the paper's judge prompt, which shows only the reference and the answer, not the question; we used one call per answer at temperature 0, and no call failed. LAVE is our main Bangla metric because EM is close to zero for all models. We also checked LAVE with two rule-based audits that use no judge: whether a yes/no answer is correct in either language, and whether the modality named in a modality answer is correct.

Each medical model is compared with its base model on the same items. We report the paired difference with a 95% bootstrap interval (5,000 resamples of items), and for accuracy the exact McNemar test [28].

4. RESULTS AND ANALYSIS

4.1. Main result: medical versus base model

Table 3 compares each medical model with its base model on all three tasks. Medical adaptation did not give a consistent advantage.

MedVLM-R1 behaved like a copy of Qwen2-VL. It gave identical answers on every BUS-BRA image, on 93% of RadImageNet-VQA questions and on 72% of BanglaMedVQA questions, and none of the differences is distinguishable from zero. This fits its training: it was tuned on 600 MRI multiple-choice questions to write a reasoning block before a letter answer. Our prompts asked for a bare answer, the model complied, and the reinforcement learning left its direct answers essentially unchanged.

MedGemma changed its answers a lot, but not consistently for the better. Only 35–55% of its answers matched Gemma 3's. It was clearly better on Bangla, ahead but not significantly on BUS-BRA, and slightly (not significantly) behind on RadImageNet-VQA. The next three subsections show what each task adds to this picture.

4.2. BUS-BRA: accuracy hides constant answers

Three of the four models scored at chance on BUS-BRA (Table 4), and the reason is visible in their answers: Qwen2-VL and MedVLM-R1 answered "benign" for all 400 images, and Gemma 3 answered "malignant" for 391. On a balanced sample a constant answer earns exactly 50%. Gemma 3's 98% sensitivity therefore means nothing on its own: with 2.5% specificity it flags almost every patient.

MedGemma was the only model whose answers depended on the image. It reached 57.5% (95% CI 52.6–62.3%), significantly above chance ($p = 0.0016$, still significant after Bonferroni correction for four models). It leaned towards "benign", however, and missed 118 of the 200 cancers. Its gain over Gemma 3 (+29 images) was not significant in the paired test (144 images correct only for MedGemma against 115 only for Gemma 3). So MedGemma shows a weak signal that its base model lacks, but this sample does not establish that medical training caused it.

Table 2: Top: the prompts (verbatim; line breaks shown as ↩). The image is placed before the text in one user turn. Bottom: one example question from our samples for each question type. RadImageNet-VQA options are not reproduced (our result files store only the question text). English glosses for BanglaMedVQA are the dataset's own English versions; the models saw only the Bangla.

Prompts	
BUS-BRA	`You are a medical assistant. Examine this breast ultrasound image.↩ Classify the lesion as exactly ONE of the following:↩ - benign: non-cancerous lesion↩ - malignant: cancerous lesion↩↩ Respond with ONLY the label: benign or malignant.`
RadImageNet-VQA, MCQ	`You are a medical assistant. Look at this image and answer the following multiple-choice question with ONLY the letter of the correct option (e.g. 'B'), nothing else.↩↩ Question: {question}↩ Options:↩ A. {A}↩ B. {B}↩ C. {C}↩ D. {D}`
RadImageNet-VQA, yes/no	`You are a medical assistant. Look at this image and answer the following question with ONLY the single word 'Yes' or 'No'.↩↩ Question: {question}`
BanglaMedVQA (system prompt of [11]; user turn = Bangla question)	`You are a careful, clinically grounded medical vision-language assistant. Task: You will be given a medical image and a single question about that image (e.g., modality, organ, abnormality, condition, or the position of the condition). Your job is to look at the image and provide the exact, minimal answer to the question. CRITICAL: You must respond with ONLY the exact answer. No explanations, no sentences, no extra words. STRICT OUTPUT RULES: - ONE word or short phrase only - NO sentences or explanations - NO "This is..." or "The image shows..." - NO punctuation unless part of the answer`

Example questions		**Reference answer**	**Chance**
RadImageNet-VQA (CT/MRI), 100 questions per group			
Anatomy, MCQ	Which of the following anatomical regions is depicted in this scan?	one letter (A–D)	25%
Anatomy, yes/no	Is the knee visible in the image?	yes	50%
Abnormality, yes/no	Does the lung look abnormal?	no	50%
Pathology, yes/no	Is the abdomen affected by bowel mass?	yes	50%
Pathology, MCQ	What is the most likely pathology in this image?	one letter (A–D)	25%
BanglaMedVQA (Bangla), 50 questions per category and source			
Modality	এই গবেষণায় কোন ইমেজিং পদ্ধতি ব্যবহার করা হয়েছে? (What imaging modality is used in this study?)	এক্স-রে (X-ray)	–
Organ	ছবিতে কোন অঙ্গটি দেখানো হয়েছে? (What organ is depicted in the image?)	মস্তিষ্ক (brain)	–
Abnormality	কোনও অস্বাভাবিকতা আছে কি? (Are any abnormalities present?)	হ্যাঁ (yes)	50%
Condition	ছবিতে কী কী অবস্থা দেখা যাচ্ছে? (What conditions are observed in the image?)	দুটি পাথর, এন্ডোক্লিপ (two stones, endoclip)	–
Position	নডিউলটি কোথায় অবস্থিত? (Where is the nodule located?)	ডানদিকের মাঝখানে (middle right)	–

Table 3: Medical model minus its base model on the same items: difference in accuracy (BUS-BRA, RadImageNet-VQA) or LAVE (BanglaMedVQA), in percentage points, with 95% paired bootstrap interval; exact McNemar p; share of identical answers.

	MedVLM-R1 − Qwen2-VL	**MedGemma − Gemma 3**
BUS-BRA	0.0 identical answers on all 400	+7.3 (−0.3, 15.0) $p = 0.08$; 35% identical
RadImageNet-VQA	+0.8 (−1.2, 2.8) $p = 0.57$; 93% identical	−1.8 (−7.2, 3.6) $p = 0.56$; 55% identical
BanglaMedVQA	+0.8 (−0.3, 2.0) 72% identical	**+7.4** (4.8, 10.0) 36% identical

4.3. RadImageNet-VQA: anatomy is easy, disease is not

Overall accuracy was similar for all four models: 64.2% (Qwen2-VL), 65.0% (MedVLM-R1), 63.0% (Gemma 3) and 61.2% (MedGemma), against a chance level of 40%. The question groups tell a clearer story (Fig. 3). Every model recognised anatomy well (79–88% on both anatomy groups), but on questions about disease all of them were near chance: 52–61% on abnormality and pathology yes/no (chance 50%) and 32–37% on pathology multiple choice (chance 25%). With 100 questions per group the 95% interval is about ±9 points, so several of the disease scores cannot be separated from guessing. Neither medical model narrowed this gap; MedGemma was in fact below Gemma 3 on both anatomy groups, although neither difference is significant ($p = 0.39$ and 0.79).

Table 4: BUS-BRA results (400 images, 200 per class). "Malig." = number of "malignant" answers; Sens./Spec. = sensitivity/specificity for malignant (%); p: one-sided binomial test against 50%.

Model	**Acc. % (95% CI)**	**Malig.**	**Sens.**	**Spec.**	p
Qwen2-VL-2B	50.0 (45.1–54.9)	0	0.0	100.0	0.52
MedVLM-R1	50.0 (45.1–54.9)	0	0.0	100.0	0.52
Gemma 3 4B	50.3 (45.4–55.1)	391	98.0	2.5	0.48
MedGemma 4B	**57.5** (52.6–62.3)	134	41.0	74.0	0.002

When the image does not decide, the models fall back on preferences (Table 5). Because each yes/no group was balanced, an unbiased model would say "yes" about half the time. Gemma 3 said "yes" to 74% of yes/no questions, including 98 of the 100 abnormality questions. MedGemma, trained from it, leaned the other way, with 42% "yes" overall and 32% on pathology. Medical training did not remove the bias; it reversed it, and on abnormality questions both models ended up near chance from opposite directions. On pathology multiple choice the correct letters were spread evenly (A 25, B 22, C 25, D 28), yet each pair favoured one letter: B for the Gemma pair, C for the Qwen pair. This matches the option-letter bias known for language models [22] and suggests the preference is inherited from the base model.

The four models also failed together: all of them were

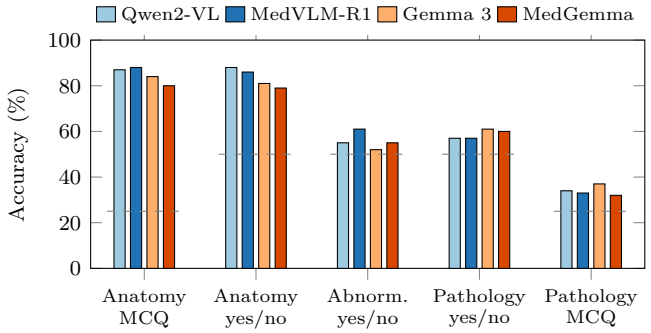


Figure 3: RadImageNet-VQA accuracy by question group (100 questions each). Dashed lines mark chance. Light bars are general models, dark bars their medical versions.

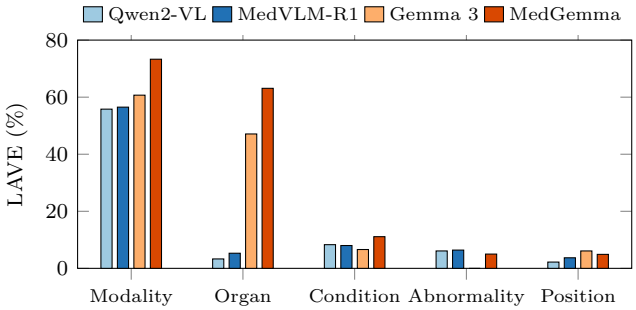


Figure 4: BanglaMedVQA LAVE by question category, averaged over the chest X-ray and MedICaT sources (100 questions per category).

Table 5: RadImageNet-VQA answer preferences. Share of "yes" answers per yes/no group (true share 50%), and the most chosen letter on pathology MCQ with its count (the correct letter was B in 22 of 100 questions, C in 25).

	Qwen2	MedVLM	Gemma3	MedGemma
"Yes", anatomy (%)	42	42	49	55
"Yes", abnormality (%)	77	69	98	39
"Yes", pathology (%)	43	47	75	32
"Yes", all yes/no (%)	54	53	74	42
Top letter, path. MCQ	C (37)	C (35)	B (35)	B (42)

wrong on the same 65 questions (13%). Accuracy varied by body region (hip was hardest for every model, 37–53% on 38 questions), but most region-level cells are too small to interpret.

4.4. BanglaMedVQA: the one clear gain, and why it needs care

BanglaMedVQA is the only task where a medical model clearly beat its base model (Table 6): MedGemma reached 31.5% LAVE against 24.1% for Gemma 3. The two Qwen-family models were close to each other at about 15–16%. Exact match was near zero for all models, which is why LAVE carries the comparison.

The gain is in the easy categories. Fig. 4 shows the same anatomy-versus-disease pattern as RadImageNet-VQA. Modality was the easiest category for every model, and organ questions were answered well only by the Gemma pair. Condition, abnormality and position stayed between 0% and 13% LAVE for all four models in both sources. MedGemma's lead over Gemma 3 came almost entirely from chest X-ray organ (+30.6 points) and chest X-ray modality (+20.4) questions, the only two cells whose paired intervals exclude zero. Modality is also close to a base-rate question here: X-ray is the reference for 56 of our 100 modality questions.

The models mostly answered in English. Although every question was in Bangla, 82–98% of each model's answers contained no Bangla script; MedGemma, the best model, answered in Bangla least often. When the Qwen models did answer in Bangla the answer was often meaningless; their most frequent answer to chest X-ray organ questions was লাইন ("line"). They also answered some MedICaT questions with a bare option letter although no options were given (39 answers for Qwen2-VL, 27 for MedVLM-R1). Several models repeated one stock answer: MedVLM-R1 said "pneumonia" to 37 of the 50 chest X-ray condition questions. On the balanced abnormality questions, the yes/no preferences from RadImageNet-VQA appeared again: the Qwen models never answered "yes", while 62 of MedGemma's 79 clear yes/no answers were "yes".

The metric under-credits correct English answers (Table 7). The LAVE judge sees only the Bangla reference and the model's answer. MedGemma answered 34 Bangla "yes" questions with the English "Yes"; every one of these correct answers scored 0, while its three answers of হ্যাঁ (Bangla "yes") to the same kind of question all scored 1.0. The judge was also inconsistent across models: some of the Qwen models' correct "no" answers received credit, but none of the Gemma models' correct yes/no answers did. Correct "CT" answers to modality questions averaged about 0.31 LAVE against about 0.87 for correct "X-ray" answers, for every model. BERTScore without rescaling did not help either: incorrect answers still scored about 66–70, and its rank correlation with LAVE was only 0.27–0.43.

So the LAVE scores in Table 6 are a noisy lower bound, and the size of the gap between models partly depends on which language they answer in. This also limits comparison with the original paper, which reported 19.85% zero-shot LAVE for MedGemma 4B on the test split [11], against our 31.5% on a different sample. Table 8 shows typical answers.

Table 6: BanglaMedVQA results (500 questions). EM = exact match (%). LAVE with 95% bootstrap interval. "No Bangla" = share of answers containing no Bangla script (%).

Model	EM	BERTScore	LAVE (95% CI)	No Bangla
Qwen2-VL-2B	0.0	68.6	15.1 (12.5–17.8)	82.0
MedVLM-R1	0.0	68.4	16.0 (13.4–18.7)	89.2
Gemma 3 4B	2.4	69.0	24.1 (20.9–27.3)	90.0
MedGemma 4B	1.6	**71.4**	**31.5** (28.1–34.9)	98.2

Table 7: Audit of LAVE on answers that are correct by a rule-based check. Top: correct yes/no answers to abnormality questions and how many the judge credited (LAVE > 0). Bottom: correct modality answers found by keyword matching (of 100).

	Qwen2	MedVLM	Gemma3	MedGemma
Correct yes/no answers	31	34	12	53
credited by LAVE	8	11	0	0
Correct modality (rule)	82	84	98	99
Modality LAVE (%)	55.8	56.5	60.7	73.3

4.5. Discussion

Across the three tasks the four models behaved similarly. They recognised *what kind of image* they were given (modality, anatomy) much better than *what was wrong* in it. When the image did not settle the answer, they used a fixed preference (one class label, "yes" or "no", one option letter, one stock disease name), and this preference was either kept or simply reversed by medical training. In Bangla they mostly answered in English, which the standard metrics do not score fairly.

Each task contributes a different piece of evidence to the main finding. BUS-BRA shows that the reinforcement-learning adaptation did not change behaviour at all, and that MedGemma's improvement over a chance-level base is weak and not significant. RadImageNet-VQA shows that neither adaptation closed the gap between anatomy and disease, even though MedVLM-R1 was trained on MRI questions of a similar style. BanglaMedVQA gives the only clear gain, but it comes from general questions, and its size depends on how answers in the wrong language are scored.

Two practical points follow. First, balanced samples and answer distributions were essential: on BUS-BRA's natural class mix (68% benign) the always-"benign" Qwen models would have reached about 68% accuracy, and on the full RadImageNet-VQA benchmark, where 72% of abnormality questions have the answer "yes", Gemma 3's "yes" preference would look like skill. Reporting the share of each answer costs nothing and exposed every such case. Second, a medical model should be compared with its own base model before claiming an improvement, as Jeong et al. [5] also argue. For the small models we tested, general medical training was not enough to improve zero-shot diagnosis; this points to the need for training methods and high-quality data aimed specifically at diagnostic questions and at low-resource languages.

5. LIMITATIONS

One shared prompt per task. An identical prompt isolates the effect of the weights, but does not give each model its best conditions. MedVLM-R1 was trained to reason in `<think>` tags before answering (its model card generates up to 1,024 tokens); we tested it only in direct-answer mode, so we cannot say whether its reasoning mode helps. MedGemma might also do better with its own recommended prompts.

Small samples and one run. Each RadImageNet-VQA group has 100 questions and each BanglaMedVQA cell 50, so group-level intervals are wide. Each item was run once with greedy decoding.

Text outputs only, no image-ablation control. We stored decoded answers, not probabilities, so we cannot compute AUC on BUS-BRA; a model with a strong label preference might still rank lesions well. We also did not run blank-image controls, which would show how much each answer depends on the image.

Bangla evaluation. LAVE relies on a proprietary judge that sees no question and scored correct English answers inconsistently, and BERTScore was computed without rescaling. The BanglaMedVQA references were LLM-generated and machine-translated, with expert checks on a subset only. We did not run the English versions of the questions, which would separate visual errors from language errors, and our train-split sample differs from the original paper's test split.

Possible training-data overlap. MedVLM-R1's training questions come from the OmniMedVQA collection; example questions on its model card use labels such as "flexor pathology" that also appear in our RadImageNet-VQA sample. MedGemma's training data are not fully public. Overlap, if any, would favour the medical models.

Reproducibility and scope. Model revisions and library versions were pinned for BUS-BRA but not recorded for the two VQA tasks. We tested two pairs of 2B and 4B models on three datasets; larger models or fine-tuned use may behave differently. None of these results is a clinical evaluation.

6. CONCLUSIONS

We compared two small medical VLMs with the general-purpose models they were built from, under identical zero-shot conditions, on breast ultrasound classification, closed-ended CT/MRI question answering and open-ended Bangla question answering. Medical adaptation did not consistently improve performance: MedVLM-R1 was nearly indistinguishable from Qwen2-VL, and MedGemma improved on Gemma 3 only in Bangla, mainly on modality and organ questions. All four models recognised anatomy far better than disease, fell back on fixed answer preferences, and mostly answered Bangla questions in English. Future work should compare medical models with their base models, report answer distributions alongside accuracy, test each model in its intended prompting mode, and develop training methods and data aimed at diagnostic reasoning and low-resource languages.

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Table 8: Example BanglaMedVQA answers from our sample, with each answer's LAVE score in brackets. English glosses are for the reader. The second row is the same kind of question as the first, answered correctly in English by MedGemma and scored 0.

Source / type	Question	Reference	Qwen2-VL	MedVLM-R1	Gemma 3	MedGemma
Chest X-ray, abnormality	কোনও অস্বাভাবিকতা আছে কি? (Any abnormality?)	হ্যাঁ (yes)	no [0.0]	no [0.0]	Consolidation [0.0]	হ্যাঁ [1.0]
MedICaT, abnormality	ছবিতে কি কোনো অস্বাভাবিকতা দেখা যাচ্ছে? (Is an abnormality visible?)	হ্যাঁ (yes)	A: A mass in the right temporal lobe [0.0]	A: There is no abnormality [0.0]	Lesions [0.0]	Yes [0.0]
MedICaT, organ	ছবিতে কোন অঙ্গটি দেখানো হয়েছে? (Which organ?)	মস্তিষ্ক (brain)	A [0.0]	A [0.0]	মস্তিষ্কের [0.8]	মস্তিষ্ক [1.0]
Chest X-ray, position	নডিউলটি কোথায় অবস্থিত? (Where is the nodule?)	ডানদিকের মাঝখানে (middle right)	lung [0.0]	lung [0.0]	ডান ফুসফুস (right lung) [0.3]	Lung [0.0]
MedICaT, position	(Where is the diffusely hyperechoic myocardium with acoustic shadowing?)	বাম ভেন্ট্রিকল (left ventricle)	C [0.0]	A [0.0]	Left ventricle [0.9]	Heart [0.0]

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